Watt Window

Advisory scheduling for noncritical loads

A cheap time slot can conflict with another load or a deadline.

Help a household or small shop compare an entered schedule with a lower-cost plan that respects an explicit capacity model.

Make the assumptions inspectable

ENTERED INPUTS

Tariff per time slot
Estimated energy and concurrent power
Time windows, duration and fixed starts
Background load and capacity

BOUNDED SEARCH

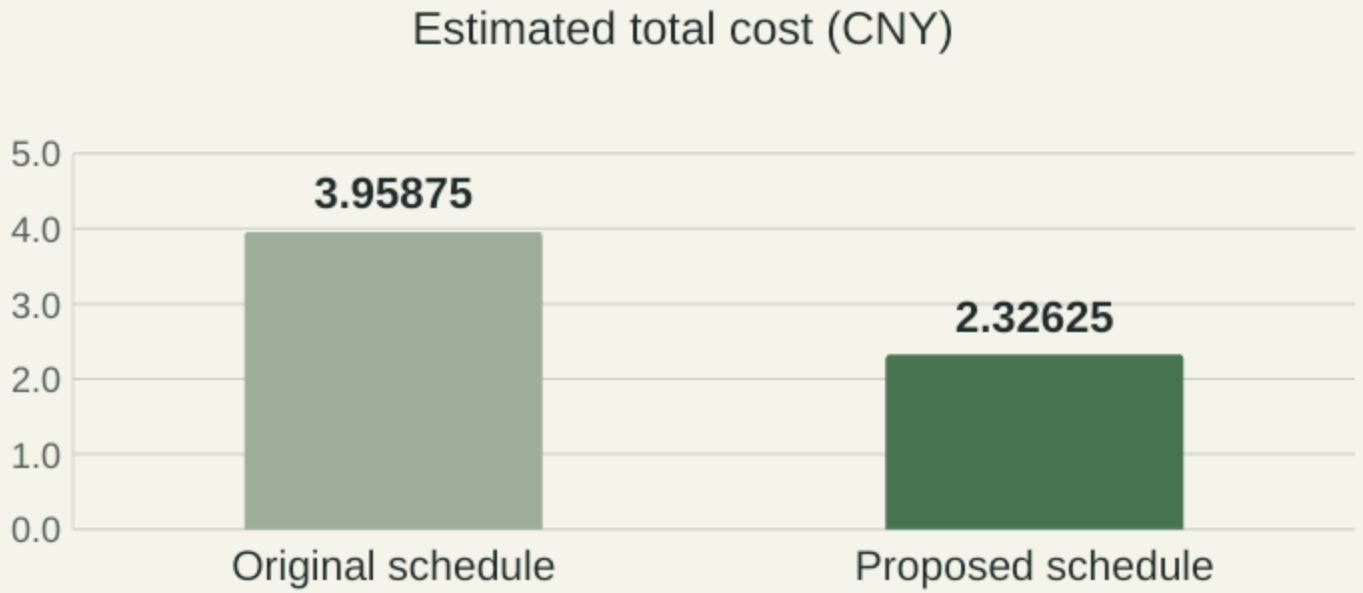
Contiguous jobs on a finite grid
Exact rational cost comparison
Branch-and-bound with a node budget
Independent check of each returned plan

The result says what was proven

Optimal in this model · Feasible, not proven optimal · No plan within budget · Proven infeasible

A lower estimated bill, under fictional inputs

Six-hour household example · CNY, including background load



1.63250 CNY

Estimated model cost difference

12 × 30-minute slots
3 noncritical jobs
2,600 W entered capacity
350 W background load

Fictional tariffs and load estimates; no field trial. Estimated energy is unchanged (4.3 kWh). Proposed peak rises from 1,850 W to 2,250 W, within the entered limit.

A narrow, reproducible advisory prototype

Project / license	Established capability	Scope of this prototype
EMHASS / MIT	Residential optimization, solar, storage and Home Assistant	Offline, manually entered noncritical jobs; no device integration
FlexMeasures / Apache-2.0	Multi-asset scheduling, time series, forecasts and API/UI	Small finite-grid model with standard-library Python
Watt Window / MIT	Independent implementation; no upstream code copied	Proof status, baseline checks and inspectable exports

Evidence available now

17 automated tests, including an independent brute-force oracle. An overload witness rejects 4,800 watt-slots of demand against 3,300 available. A one-node run reports an unproven feasible plan.

A proposed 48-hour build and evaluation

Time in event	Planned work	Reviewable deliverable
0–8 hours	Validate scope and new assumptions	Dated input protocol and test cases
8–24 hours	Evaluate new constraint / failure cases	Reproducible runs and oracle comparisons
24–40 hours	Improve explanations and manual review	Usability findings and revised exports
40–48 hours	Re-run checks and prepare final pitch	Versioned delta, limitations and demo

If shortlisted: 28 November 09:00 EST → 30 November 09:00 EST

Team and first-stage readiness

TEAM

Solo project

Author details pending confirmation

PREPARED FOR REVIEW

Working offline prototype and examples
MIT-licensed original code
Reproducible model results and tests
Editable first-stage presentation draft

Before submission

Confirm author and eligibility details, review the pre-event baseline with the organizer, and complete the official application. Registration and shortlisting are not yet claimed.